- In today’s digital world, computer systems and networks are at the heart of nearly every activity—from banking, education, and healthcare to government services and personal communication.
- These systems hold sensitive information such as financial records, exam results, medical histories, and even our personal identities.
- Because of this reliance, protecting computers and the data they store has become critical. Without proper security, individuals, businesses, and governments face risks

Key definitions

- **Computer System Security:** The protection of computer systems and networks from theft, damage, disruption, or unauthorized access. It encompasses measures to safeguard hardware, software, data, and network components.
- **Information Security:** The practice of protecting information from unauthorized access, disclosure, modification, destruction, or disruption.
- **Computer Security Threat/Risk:** Any event or action that can harm or damage computer hardware, software, or data.
- **Security control** refers to the measures and procedures put in place to protect information systems and data from unauthorized access and security threats.

REASONS WHY COMPUTER SYSTEMS ARE VULNERABLE

Digital data stored on computers and servers is exposed to more threats compared to data stored on paper.

- **Outdated Software:** Old software that isn't updated can be easily exploited by attackers.
- **Lack of Training:** Employees without cybersecurity training may unintentionally cause security breaches.
- **Weak Passwords:** Poor password practices, like using weak or repeated passwords, make systems easier to hack.
- **No Response Plan:** Without a plan for handling security incidents, organizations struggle to manage and recover from attacks.
- **Improper Security Settings:** Incorrect settings can leave systems open to attacks.
- **Cyber Threats:** Hackers and cybercriminals may target systems to steal, disrupt, or damage data.
- **Dishonest Employees:** Insider threats from employees with bad intentions can also compromise security.

PHYSICAL THREATS AND CONTROLS

- Physical Threats: cause physical damage/loss to computer systems.
- Physical Controls: security measures implemented to protect against physical threats.

Physical Threat	Physical Control
Natural Disasters – Floods, earthquakes, fires, storms	fire suppression, climate control, water detection systems
Theft – Stealing of computer equipment.	locks, security cards, biometrics, guards
Vandalism – Defacing or destroying equipment like servers or storage devices.	CCTV cameras, alarms, monitoring systems
Terrorism – Sabotage or violence targeting data centers or communication networks.	CCTV cameras, alarms, monitoring systems
Power Failures – Unexpected power outages damaging hardware & causing data corruption.	Backup Power Systems (UPS, standby generators)
Unauthorized Physical Access – Individuals entering server rooms or restricted areas to tamper/steal information.	biometric systems, smart cards, locks, security personnel
Old/Unused Equipment – Sensitive data leakage from discarded devices.	Secure Hardware Disposal (data wiping, physical destruction)

LOGICAL THREATS AND CONTROLS

- **Logical Threats:** non-physical threat that targets information systems or data through unauthorized access, manipulation, or malicious software.
- **Logical controls:** measures to prevent unauthorised individuals from gaining access to data, software, computer operating systems, networks and other computer equipment, when they have managed to gain physical access.
- Logical threats are categorized into;
  - Accidental
  - Intentional

LOGICAL THREATS AND CONTROLS

A. ACCIDENTAL THREATS

- **Human Errors:** Mistakes made by individuals while performing their tasks that can lead to security vulnerabilities, data loss, or system disruptions. For example, accidentally deleting important files or data and entering incorrect data into a system (e.g., wrong financial information).

Controls

- **Training and Awareness**: Regularly train employees on security protocols.
- **Clear Documentation**: Maintain up-to-date documentation on procedures and policies. This should be easily accessible and clear to prevent misunderstandings and mistakes.

LOGICAL THREATS AND CONTROLS

- **Procedural Errors:** occur when system installation protocols are not followed. These errors can cause problems like data breaches or system failures.

Controls

- **Standard Operating Procedures (SOPs)**
  - ✔ Create clear and detailed steps for important tasks, like setting up software or handling data.
  - ✔ Make sure everyone follows these steps every time.
- **Regular Audits**
  - ✔ Check and review the rules and steps regularly to make sure they still work well.
  - ✔ Update them if needed to fix any mistakes or keep up with changes in technology.

LOGICAL THREATS AND CONTROLS

- **Software Errors:** flaws in software that can cause systems to behave unexpectedly or fail altogether. For example, Software crashes due to compatibility issues.

Controls

  - **Regular Updates and Patches**: Keep all software up-to-date with the latest patches and updates to fix known bugs and vulnerabilities.
  - **Testing and Validation**: Thoroughly test new software before deploying to ensure compatibility with existing systems and software.

- **Electromechanical Problems:** Issues that arise from the malfunction or failure of physical components, such as hardware. For example, Hard drive failures leading to data loss.

Control

  - **Preventive Maintenance**: Perform regular maintenance checks on hardware components to identify and address potential issues before they lead to failure.
  - **Monitoring Tools**: Use monitoring tools to keep an eye on hardware performance and health.

LOGICAL THREATS AND CONTROLS

B. INTENTIONAL THREATS

- **Computer Viruses**

Controls

  - Regularly update antivirus software on all systems.
  - Use email filtering to block malicious attachments and links.
  - Educate users about the dangers of downloading and opening unknown files.

- **Computer fraud**

The use of computers or digital systems to commit fraud, such as identity theft, financial fraud, or manipulating data for personal gain.

Controls

  - Limit access to sensitive data and financial systems to only those employees who need it for their job.

CATEGORIES OF CONTROLS

A. GENERAL CONTROLS

Focus on maintaining the overall security of the information systems

- **Physical Controls**: These are measures to prevent physical access to systems and infrastructure.
- **Logical Controls**: These are measures to prevent unauthorized access to systems and data.
- **Administrative Controls**: encompass a range of policies, procedures, and practices designed to manage and mitigate risks. i.e
  - Policies which define who can access what information and under what conditions
  - Procedures for adding and removing users from systems

CATEGORIES OF CONTROLS

B. APPLICATION CONTROLS

Automated checks and procedures built into software applications to ensure accuracy, completeness, validity, and security of data throughout the input–processing–output cycle.

- **Input Controls**
  - Ensure data entered is accurate and complete. For instance, Student ID must follow pattern MCA/2025/001; wrong format will be rejected by the system.
  - Ensuring the correct type of data is entered (e.g., numeric, text). For example, marks entered must be between 0 and 100. If a lecturer enters 150, the system rejects.

CATEGORIES OF CONTROLS

B. APPLICATION CONTROLS

- **Processing Controls**
  - Ensure data is processed correctly and consistently.
  - Ensures that detailed logs of all transactions is recorded for auditing purposes. For instance**,** If someone edits exam marks, the system records "X changed Mark from 70 → 75 on 15/08/2025".

- **Output Controls**
  - Ensure output is accurate and reaches the correct recipients. i.e. Students log in with their ID to see only their own results.
  - Restrict access to sensitive output data and log who receives it.

CYBER SECURITY

- **Cyberattack:** an assault launched by cyber criminals against a single or multiple computers or networks.

  Examples of cyberattacks:
    - Hacking
    - Viruses
- **Cyber security:** controls taken to reduce the probability of being the victim of a cyberattack, and procedures to recover if a cyberattack takes place.
- Given the potential consequences of a cyberattack, it is important to have procedures in place to manage the risk of cyberattacks.

Importance of Cybersecurity

- **Protecting our Digital Lives** – From your online banking to your social media, your personal information is valuable. Cyberattacks can steal your identity, money, or even ruin your reputation.
- **Safeguarding Business** – Companies rely on computers and networks to operate. A cyberattack can cause financial loss, damage reputation, and even put people's jobs at risk.

etc

Challenges to Cybersecurity

While Cybersecurity has a lot of features and advantages, it also has some challenges;

- **Cost and Time** – Implementing and maintaining cybersecurity systems can be expensive and time-consuming for organizations.
- **Complexity** – Cybersecurity systems can be complex to manage and maintain in the longer run.
- **Constant Evolution** – Cyber threats are constantly evolving, which require the Cybersecurity solution to be consistently adaptive to these threats.
- **Limited Effectiveness** – Despite maximum efforts, complete protection from cyberattacks is impossible to attain.

Cybersecurity Controls in E-commerce

- **E-commerce** platforms are prime targets for cybercriminals due to the sensitive financial and personal data they handle.
- The following slides explain some of the control measures that protect e-commerce platforms from cyberattacks.

Cybersecurity Controls in E-commerce

FIREWALLS

- A firewall is a computer network security system that is located between the network and the internet. The firewall is configured to inspect network traffic that passes between the network and the internet. There are two types of firewalls:
  - **Hardware firewalls** are typically included within broadband routers. They protect the computers that they are connected to.
  - **Software firewalls** run on the personal computer and can be customised by the user themselves.

Advantages of firewalls

- ✔ Protects the computer from unauthorised access.
- ✔ Firewalls can block websites and emails from suspicious sources.

Disadvantages of firewalls

- ✔ Firewalls cannot prevent the spread of malware that is introduced through pen drives or other periphery devices.
- ✔ Firewalls cannot prevent attacks by insiders.

Hardware and software firewalls

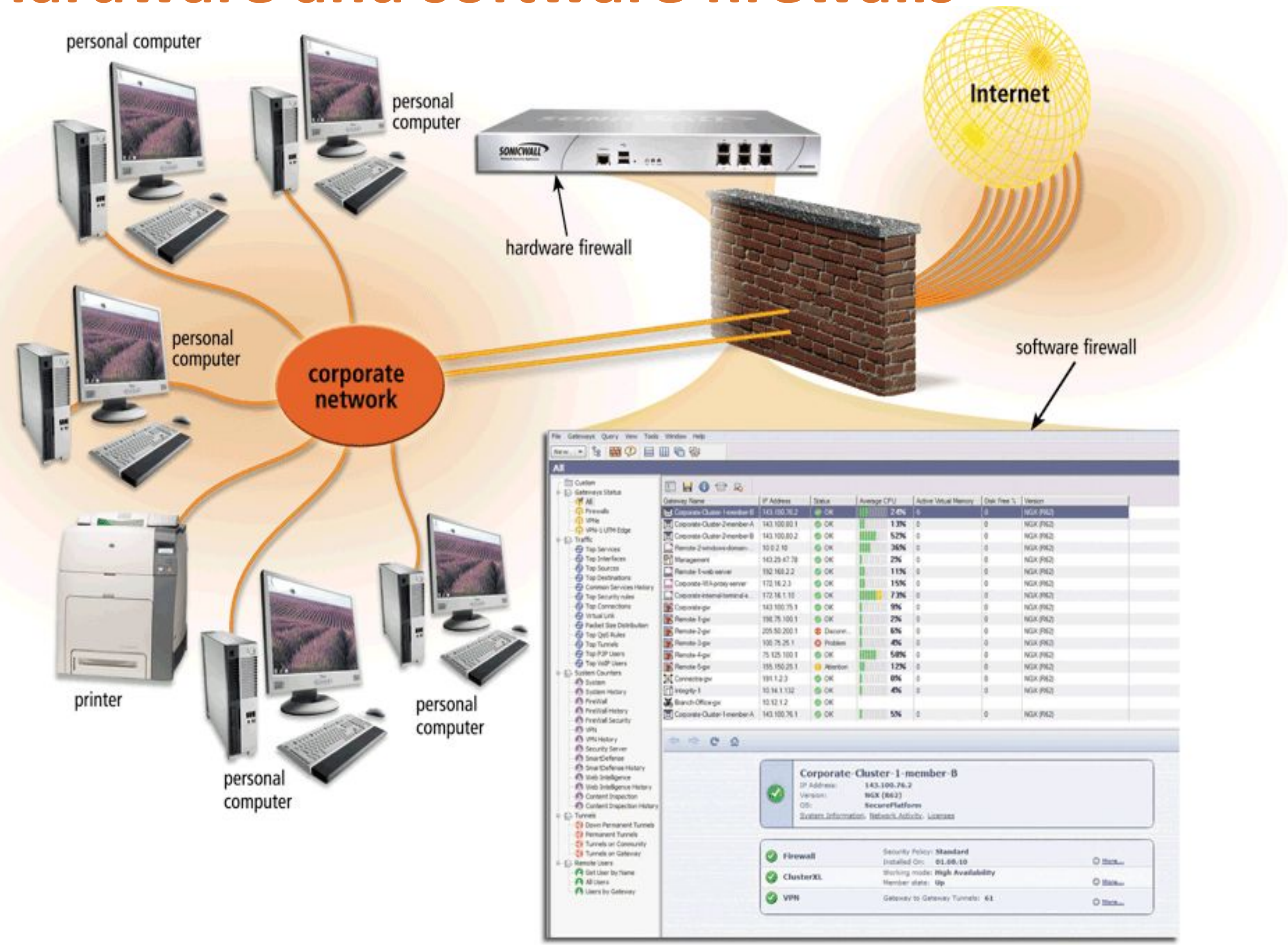

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Figure 11-8

Cybersecurity Controls in E-commerce

ENCRYPTION

- A process of converting readable data into unreadable characters to prevent unauthorized access. Encryption ensures that data traveling along networks cannot be read by unauthorized users.
- Readable data is called plaintext & encrypted data is ciphertext. An encryption key is a set of characters that the originator of the data uses to encrypt the plaintext and the recipient of the data uses to decrypt the ciphertext.

  Two methods of encryption

  - Symmetric key encryption: both the originator and the recipient use the same secret key(private key) to encrypt and decrypt the data.
  - Asymmetric key encryption: uses two encryption keys: a public key and a private key.

Cybersecurity Controls in E-commerce

AUTHENTICATION

- The process of verifying the identity of a user, system, or entity to ensure that they are who they claim to be.
- Authentication ensures that only authorized individuals can access accounts, make purchases, or manage the online store.
- Many systems use **two-factor authentication** as a more secure method of giving access to systems. When a user has entered a password, a code is then sent to a registered smartphone or landline number owned by the user. The user is required to enter this code to gain access.

Cybersecurity Controls in E-commerce

DIGITAL SIGNATURE

- A digital signature is a cryptographic method that ensures the authenticity and integrity of a digital message or document.
- Digital signatures help to prevent e-mail forgery.
- It serves as an electronic equivalent of a handwritten signature but is far more secure.
- It is attached to an electronic message to verify the identity of the message sender.

Cybersecurity Controls in E-commerce

DIGITAL CERTIFICATES

- Data file used to establish the identity of users and electronic assets for protection of online transactions.
- E-commerce platforms commonly use digital certificates. Web browsers, such as Internet Explorer, often display a warning message if a Web site does not have a valid digital certificate.
- A certificate authority (CA) is an authorized person or a company that issues and verifies digital certificates. Users apply for a digital certificate from a CA.
- A digital certificate typically contains information such as the user's name, the issuing CA's name and signature, and the serial number of the certificate.

HEALTH AND SAFETY

- Prolonged computer use can lead to various health issues if not managed properly.
- Understanding these risks and implementing appropriate control measures is essential for maintaining overall well-being and productivity of employees.
  - **Poor posture or improper workstation setup:** Consider taking breaks for some minutes to stretch and relieve muscle tension. Adjust chair and desk height to maintain a neutral posture.
  - **Eye Strain and Vision Problems:** Ensure the workspace is well-lit to reduce glare and eye strain. Adjust monitor brightness and also consider using glasses to protect your eyes from light.